

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: Bovine papilloma virus E1 in pML2d
Species Used	Mouse, C127, fibroblast, mammary tumor		

Before the Pulse

Cell Growth Medium	DMEM + 10 % Fetal Calf Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	logarithmic
		Pre-pulse Incubation	10 minutes on ice in DMEM + 10 % Fetal Calf Serum + BES
Wash Solution	PBS (Phosphate Buffered Saline) and Electroporation Medium		

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature		
Electroporation Medium*	DMEM + 10 % Fetal Calf Serum + BES	Cuvette Gap	0.4 cm
Cell Density	1.5 to 2.0 x10 ⁶ (6) cells/ pulse	Voltage	0.21 kV
Volume of Cells	0.25 ml	Field Strength	0.525 kV/cm
DNA Concentration	10 to 20 µg / pulse		
DNA Resuspension Buffer	TE (10 mM Tris, 1 mM EDTA, pH 8.0)	Capacitor	960 µF
Volume of DNA	25 µl	Resistor	(Pulse Controller) Ω none
After the Pulse		Time Constant	70 to 80 msec
Outgrowth Medium	DMEM + 10% Fetal Calf Serum		

		Relevant Publications and/or Comments Note: exponential values designated in parentheses. PBS: 1x = 8g NaCl,0.2g KCl,0.2g KH2PO4, 1.15g Na2HPO4
Outgrowth Temperature	37 °C	
Length of Incubation	3 days /transient; 2-3 weeks/ stable	
Selection Method or Assay Used	G418 resistance for stable transformation	
Electroporation Efficiency	20-30%	
Per Cent Survival	30-40%	

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193